

PRIDE DBAlign MANUAL

Multi-GNSS satellite product

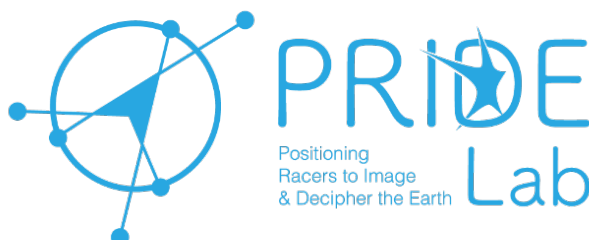
Day-Boundary discontinuity Alignment

by

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State Key Laboratory of Precision Geodesy, CAS
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**Dedicated to those
who are devoted to high-precision GNSS**



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Contents

1 Introduction	1
1.1 Acknowledgements	1
1.2 Major Features	1
1.3 Contact Us	2
2 Version modifications	3
3 Program structure and algorithm	4
3.1 Program structure	4
3.2 <i>pdba</i> command script.....	5
4 Technical Aspects	7
4.1 User Requirements	7
4.1.1 System Requirements	7
4.1.2 License	8
4.2 Installation Guide	8
4.2.1 Structures of PRIDE DBAlign	8
4.2.2 Installation and validation	9
4.3 File Specifications	10
4.3.1 Result Files.....	10
4.3.2 Usage of result data processing scripts/programs.....	13
4.4 Quick Start and Program Execution	14
4.4.1 Usage of <i>pdba</i>	14
4.4.2 Configuration File	16
4.4.3 General Operation Steps.....	17
4.4.4 Processing Example.....	17
Appendix A. Instruction on Day-Boundary Aligned Products	19

1 Introduction

1.1 Acknowledgements

PRIDE DBAlign is used for the day-boundary discontinuity alignment of GNSS precise orbit/clock/bias products, developed by the PRIDE team (<http://pride.apm.ac.cn>). We make this software package open source with the goal of promoting the implementation of United Nations (UN) International Committee on GNSS (ICG) 2025 recommendation, "Continuous GNSS Time Transfer across Day Boundaries using IGS Products" (https://www.unoosa.org/documents/pdf/icg/2025/ICG-19/ICG-19_WG-D_Recommendation_B_.pdf). In particular, PRIDE DBAlign is able to align precise satellite products across day boundaries to keep ambiguities continuous beyond the midnight epochs. As a result, GNSS positioning and time/frequency transfer spanning days could be improved by avoiding day-boundary “jumps”. PRIDE DBAlign utilizes several library functions, header files, and table files from PRIDE PPP-AR (<https://github.com/PrideLab/PRIDE-PPPAR>), another open-source software developed by the same team. The development of PRIDE DBAlign software is funded by the National Natural Science Foundation of China (Grant No. U25D8020) and supported by Sub-Commission 4.2 of the International Association of Geodesy (IAG), the Wuhan Combination Center (WCC, <https://igs.org/wg/wcc/>), and the Bias & Ambiguity Resolution Committee (BAR, <https://igs.org/wg/bias-and-ambiguity-resolution/>) of the International GNSS Service (IGS).

The open-source software PRIDE DBAlign can be downloaded at <https://github.com/PrideLab/PRIDE-DBAlign>. For the latest updates regarding support, and frequently asked questions (FAQs), please visit <http://pride.apm.ac.cn/>. The copyright of this software package is protected under the GNU General Public License (version 3). Relevant publications are

- Lin J, Geng J, Zhang Q (2025) Aligning GPS/Galileo/BDS satellite integer clock products across day boundaries for continuous time and frequency transfer. J Geod 99, 35. <https://doi.org/10.1007/s00190-025-01955-5>
- Wen Q, Geng J, Deng Y, Zhang Y (2025) Validating the IGS products in mitigating day-boundary discontinuities of kinematic positioning and time transfer. GPS Solut 29, 170. <https://doi.org/10.1007/s10291-025-01929-2>
- Geng J, Wang Y, Wen Q, Lin J, Tagliaferro G (2026) Quantifying all-frequency day-boundary discontinuities for GPS/Galileo/BDS satellite products. (*under review*)

1.2 Major Features

PRIDE DBAlign is capable of performing day-boundary alignment on multi-GNSS precise clock/bias products to restore their cross-day continuity. It can be applied to research involving cross-day precise positioning and timing, such as geodesy and time transfer. The main features of PRIDE DBAlign include:

- 1) Support for GPS, Galileo, and BDS-2/3;
- 2) Support for the alignment of precise satellite products on baseline frequencies (*e.g.*, GPS L1/L2);
- 3) Provide a DOCB module in the Bias-SINEX format for day-boundary alignment residuals;

- 4) Aligned precise products are consistent with the latest IGS conventions: the Bias-SINEX bias file format;
- 5) An additional plotting function is provided to display day-boundary alignment residuals, facilitating data analysis for researchers.

1.3 Contact Us

If you encounter any issues during the software usage, you can contact us by sending an email to report **bugs** and seek assistance.

Email: pride@apm.ac.cn

Website: <http://pride.apm.ac.cn/>



2 Version modifications

- 2026-09-18 (v1.0)
 - Release of PRIDE DBAlign v1.0



3 Program structure and algorithm

3.1 Program structure

The core program of the PRIDE DBAlign software is `algirc`. Its operational workflow is illustrated in Figure 3-1 and can be generally divided into four parts: variable definition and initialization, configuration and data reading, dual frequency precise product alignment, and output of log file and aligned products.

Variable definition and initialization. This part mainly handles the definition and initialization of variables, constants, and control parameters required for the program to run, including the satellite list, clock offset, orbit, code/phase biases, day-boundary discontinuities, and corrections. In addition, based on different GNSS systems, it initializes the frequencies, frequency ratios, combination coefficients, and wavelength parameters required for wide-lane, narrow-lane, and ionosphere-free combinations. This provides the foundational parameters for subsequent day-boundary discontinuity estimation and product alignment.

Configuration and product reading. The program first reads the configuration file and saves the product directory, SP3 precise orbit files, CLK precise clock offset files, BIA code/phase bias files, satellite list, and related control options into their corresponding variables. Subsequently, it sequentially reads the precise clock offset, precise orbit, and code/phase bias products of two adjacent days to calculate the day-boundary discontinuities of various precise products. For precise clock offset products, if the preceding day's CLK file contains the 24:00:00 epoch (i.e., the 00:00:00 epoch of the target day), it calculates the clock offset day-boundary discontinuity directly using the clock offsets of the same epoch in the two adjacent files. Otherwise, a least-squares linear extrapolation is performed using the last ten clock epochs at the end of the preceding day to extrapolate the clock offset for the preceding day at 24:00:00, provided that the extrapolation interval does not exceed 300 seconds. The extrapolated clock offset is then used to calculate the day-boundary discontinuity. When products are output, the program writes an additional CLK file for the preceding day with the “_predicted” suffix, preserving the original content and appending the extrapolated clock records after the last epoch of the original file; the target day's 00:00:00 epoch must already exist in its input CLK file and is never synthesized. For precise orbit products, the orbit files of both adjacent days are required. If the orbit products of the two adjacent days contain the required day-boundary epoch, the program directly calculates the orbit day-boundary discontinuity using the orbit positions at the boundary. Otherwise, the orbit position at the boundary is extrapolated using trigonometric-polynomial fitting and is then used to calculate the orbit day-boundary discontinuity. When orbit extrapolation is performed, the program writes an additional SP3 file with the “_predicted” suffix, preserving the original content and appending the extrapolated orbit records after the last epoch of the original file. The program projects the orbit difference at the day boundary into the satellite's radial direction and converts it into an equivalent time correction. For BIA code/phase bias products, the code and phase biases of corresponding satellites and frequencies are read across adjacent days. It calculates the day-boundary discontinuities of combined quantities such as wide-lane phase bias, narrow-lane phase bias, geometry-free bias, and code bias, providing constraints for the subsequent alignment of the integer clock and phase biases.

Dual-frequency precise clock/bias product alignment. This section is the core of the `algirc` program, primarily used to eliminate the day-boundary discontinuities in dual-frequency precise

clock/bias products of adjacent days. First, the program calculates the wide-lane phase bias day-boundary differences based on the BIA products of adjacent days and estimates the wide-lane datum correction for each satellite system to remove integer cycle jumps. Next, combining the precise clock offset day-boundary differences and the radial orbit day-boundary differences, it calculates the clock offset datum correction and constructs the day-boundary discontinuity for the Integer Recovery Clock (IRC). The program then performs an iterative estimation on the narrow-lane phase bias and IRC day-boundary differences to determine the narrow-lane datum correction and eliminate the integer narrow-lane cycle component. Finally, the wide-lane and narrow-lane corrections are converted into phase bias corrections at each original frequency, simultaneously achieving consistent correction results for clock offsets, and phase biases. Through these steps, a unified day-boundary alignment of dual-frequency precise clock offset and phase bias products is achieved.

Log file and aligned product output. After completing the dual-frequency precise product alignment, the program generates an operation log file recording the input files, processing parameters, day-boundary discontinuity calculation results, as well as clock offset, orbit, and bias corrections. This is used to verify the product alignment process and the quality of the results. Subsequently, the program generates aligned CLK and BIA products with the “_aligned” suffix. For clock offset products, the corrected satellite clock offsets of the target day are written. If the preceding day's CLK file lacks 24:00:00, an additional ‘_predicted’ CLK file for the preceding day is written as an unchanged copy with the extrapolated records appended after the last original epoch. When orbit extrapolation is performed, an additional ‘_predicted’ SP3 file is generated with the extrapolated orbit records appended after the last original epoch. For BIA products, it primarily updates the phase biases at the corresponding frequencies and can output the DOCB (Discontinuity of orbit, clock, and bias) information as needed.

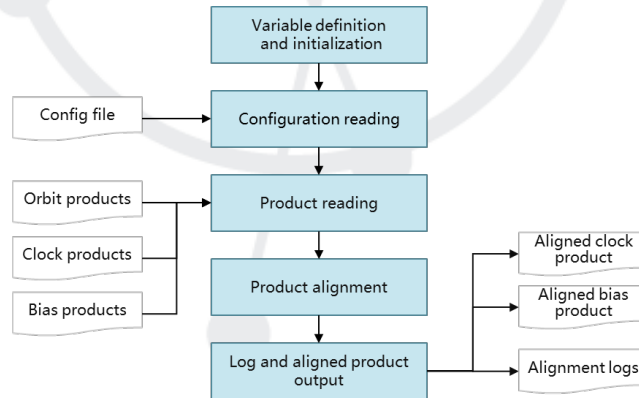


Figure 3-1 Flowchart of the algirc program

3.2 pdba command script

pdba (Product Day-Boundary Alignment) is the command-line driver script in PRIDE DBAlign for aligning GNSS precise products across a day boundary. Each invocation processes one target day and aligns its clock and bias products to the corresponding products of the preceding day. The processing date and preceding day product names are determined automatically from the input filenames. Users must ensure that the target day and preceding day products are placed in the same directory and that all command-line parameters are entered correctly.

The main functions of the current *pdba* script are described below.

- The ***main()*** function is the entry of the script, and its workflow is as follows:
 1. Parse the command-line arguments, including the clock, bias, SP3, configuration, GNSS system, write DOCB, and BDS-2/3 datum options.
 2. Check that the required clock, orbit and bias products exist and resolve their absolute paths.
 3. Calculate the preceding date and locate the corresponding preceding day clock, orbit and bias products in the same directory.
 4. Generate config_pdba_YYYYDDD in the product directory and update the product names, GNSS systems, and BDS-2/3 datum settings.
 5. Locate the algirc executable through the system PATH. Invoke algirc in the product directory to perform day-boundary alignment.
 6. Report the effective processing settings, input products, generated configuration, and completion status.
- The time-conversion functions include ***ydoy2mjd()***, ***ymd2mjd()***, ***mjd2ydoy()***, ***ydoy2ymd()***, ***gpswd2mjd()***, and ***mjd2gpswd()***:
 1. *ydoy2ymd()* converts a year and day of year into the corresponding year, month, and day.
 2. *ymd2mjd()* converts a year, month, and day into Modified Julian Date (MJD).
 3. *ydoy2mjd()* converts a year and day of year into MJD.
 4. *mjd2ydoy()* converts an MJD into the corresponding year and day of year.
 5. *gpswd2mjd()* converts a GPS week and day of week into MJD.
 6. *mjd2gpswd()* converts an MJD into the corresponding GPS week and day of week.
- The product-date processing functions include ***extract_product_date()***, ***calculate_previous_date()***, and ***previous_product_name()***:
 1. *extract_product_date()* detects and validates the date field in a product filename. It supports the long YYYYDDD[HHMM] convention and the legacy WWWWD GPS-week/day convention.
 2. *calculate_previous_date()* subtracts one day from the target date and calculates the preceding year, day of year, GPS week, and day of week.
 3. *previous_product_name()* replaces only the detected date field in the target filename to construct the expected preceding day product filename.
- The configuration-processing functions include ***set_config_value()*** and ***get_config_logical()***:
 1. *set_config_value()* updates a specified configuration item. If the item does not exist, it is appended to the generated configuration file.
 2. *get_config_logical()* reads and validates a logical configuration value such as YES or NO.
- The GNSS system functions include ***normalize_systems()***, ***filter_satellite_systems()***, and ***inspect_satellite_systems()***:
 1. *normalize_systems()* validates the value of -sys and normalizes the selected systems into the standard GEC order.
 2. *filter_satellite_systems()* comments out satellites belonging to systems that were not selected on the command line.
 3. *inspect_satellite_systems()* examines the generated configuration and determines the GNSS systems that will actually be processed.
- The output and error-handling functions include ***print_usage()***, ***print_version()***, ***error()***, ***warning()***, and ***die()***:
 1. *print_usage()* displays the command syntax, option descriptions, default values, processing rules, and examples.
 2. *print_version()* displays the current version of pdba.
 3. *error()* and *warning()* output diagnostic messages.
 4. *die()* outputs an error message and terminates the script.

4 Technical Aspects

4.1 User Requirements

4.1.1 System Requirements

PRIDE DBAlign is composed of CUI APs(Command User Interface Application Programs, CUI APs). The executable binary CUI APs included in the package support Linux and macOS. A series of tests are conducted on different operating systems with several gfortran versions. The test results are listed as in Table 4-1. Note that you can also try other Linux distribution and Fortran compiler, and tell us if you have any problems.

Fortran compiler, such as gfortran needs to be installed before installing PRIDE DBAlign.

Table 4-1 PRIDE DBAlign test results in different operating systems.

Platform version	gfortran version	Test result	Notes
Ubuntu20.04.5(x64)	11.4.0	pass	Test result is consistent with the reference
Ubuntu20.04.6(x64)	9.4.0	pass	Test result is consistent with the reference
CentOS 7 (x64)	4.8.5	pass	Test result is consistent with the reference
Debian 10(x64)	8.3.0	pass	Test result is consistent with the reference
MacOS 15.4.1	16.1.0	pass	Test result is consistent with the reference
MacOS 26.5.1	16.1.0	pass	Test result is consistent with the reference

4.1.2 License

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4.2 Installation Guide

4.2.1 Structures of PRIDE DBAlign

The structure and main files of the PRIDE DBAlign are listed in Table 4-2. We provide a complete set of cases in the “example/” folder for users to execute. To provide users with a better understanding of the software structure, further details are given below.

Table 4-2 PRIDE DBAlign structures

Directory / File	Directory / File	Instructions
bin/		Executable program
	algirc	Product alignment program
scripts/		Scripts that facilitate data process
	pdba.sh	Automatic processing Shell script
	pdba_plot.sh	Automatic plot Shell script
	plot-dcbdbd.py	Plot heatmap of DCB DBDs
	plot-ircdbd.py	Plot heatmap of NL phase bias DBDs
	plot-orbdbd.py	Plot heatmap of orbit DBDs
	plot-wlpdbd.py	Plot heatmap of WL bias DBDs
src/		Source programs

algirc/	Product alignment
header/	Header files
lib/	Library functions
Makefile	Compile Command
table/	Dependent table files
config_template	Configuration file template
sat_parameters	Satellite parameters
leap.sec	Leap seconds
install.sh	Installation script
CHANGELOG.md	Change log of the software
README.md	Software related information
LICENSE.txt	GPL3 protocol
doc/	Documentation
logo	The logo printed on screen when installing successfully
Manual	Manual of the software
example/	Examples
wum0rap/	Data of examples
test.sh	Test script

4.2.2 Installation and validation

Step 1: Make sure you have installed some essential programs in advance.

i.e. **bash**, **make**, **gfortran**

Step 2: Run script `./install.sh` to install the program automatically. (This script executes Makefile to build CUI APs and add CUI APs to system PATH (`~/.PRIDE DBAlign BIN`))

If you see the prompt shown in the picture below, the installation is completed normally.



Figure 4-1 successfully installed interface

Step 3: Input y/Y to run the example.

The script *test.sh* in `example/` folder is used to validate the correctness and effectiveness of the

installation and execution. After “test.sh” has been executed successfully, the WUM-rapid product day-boundary alignment example will be run. During data processing, progress and status information will be printed to the screen. After the run is complete, compare the generated result files with the reference results to verify that the software has been installed correctly and is functioning properly.

4.3 File Specifications

4.3.1 Result Files

After the test script has been executed successfully, the result files are generated in the “example/wum0rap/2025” and “example/wum0rap/2026” directory, as shown in Figure 4-2. The 2025 results are used as an example in the following description.

 algirc-60677.out	Alignment log
 config_pdba_2025002	Config file
 WUM0MGXRAP_20250010000_01D_01D_OSB.BIA	Previous-day bias product
 WUM0MGXRAP_20250010000_01D_05M_ORB.SP3	Previous-day orbit product
 WUM0MGXRAP_20250010000_01D_30S_CLK.CLK	Previous-day clock product
 WUM0MGXRAP_20250020000_01D_01D_OSB.BIA	Current-day bias product
 WUM0MGXRAP_20250020000_01D_01D_OSB.BIA_aligned	Aligned current-day bias product
 WUM0MGXRAP_20250020000_01D_05M_ORB.SP3	Current-day orbit product
 WUM0MGXRAP_20250020000_01D_30S_CLK.CLK	Current-day clock product
 WUM0MGXRAP_20250020000_01D_30S_CLK.CLK_aligned	Aligned current-day clock product

Figure 4-2 Result files after Product Alignment

algirc-MJD.out File

The alignment log records intermediate parameter estimates and product alignment residuals generated during the alignment procedure, thereby allowing users to assess the alignment performance. The log file comprises five sections: the file header; orbit and clock day-boundary discontinuities; dual frequency code bias and wide-lane day-boundary discontinuities; narrow-lane phase bias day-boundary discontinuities; and alignment corrections for code and phase biases.

The header format is illustrated below. It reports the executable name, execution time, working directory, and product file names.

```

1  %% INTEGER-RECOVERED CLOCK DAY-BOUNDARY DISCONTINUITY ANALYSIS FILE
2  %% Executor:                algirc
3  %% Execute Time:            2026-07-15 22:44:31
4  %% Working Directory:       /home/wyy/code/PRIDE-DBAlign/example/wum0rap
5  %% Input:                   /home/wyy/code/PRIDE-DBAlign/example/wum0rap/align_run/2025/config_2025002_tmp
6  %% SP3 Files:               WUM0MGXRAP_20250010000_01D_05M_ORB.SP3 WUM0MGXRAP_20250020000_01D_05M_ORB.SP3
7  %% CLK Files:               WUM0MGXRAP_20250010000_01D_30S_CLK.CLK_aligned WUM0MGXRAP_20250020000_01D_30S_CLK.CLK
8  %% OSB Files:               WUM0MGXRAP_20250010000_01D_01D_OSB.BIA_aligned WUM0MGXRAP_20250020000_01D_01D_OSB.BIA
9

```

Figure 4-3 Header of the Alignment Log File

As illustrated below, this section reports the radial-orbit and clock day-boundary discontinuities in millimeters and picoseconds, respectively, together with the clock adjustment applied at the day boundary.

10	1. RESULTS OF RADIAL ORBIT CORRECTION AND CLOCK									
11										
12	PRN	FRQ	A	MJDAY	RADORB	CLOCK	RADORB	CLOCK	CLK_RMS	CLK_CORR
13					(mm)	(mm)	(ps)	(ps)	(ps)	(ps)
14										
15	G02	0		60677	-21	-113	-71	-376	0	212
16	G03	0		60677	-14	-73	-48	-243	0	212
17	G04	0		60677	3	-76	9	-254	0	212
18	G05	0		60677	-31	-73	-103	-244	0	212
19	G06	0		60677	6	-62	21	-206	0	212
20	G07	0		60677	18	-43	59	-144	0	212
21	G08	0		60677	19	-80	62	-267	0	212
22	G09	0		60677	-19	-90	-63	-299	0	212
23	G10	0		60677	-10	-85	-32	-285	0	212
24	G11	0		60677	-11	-79	-37	-264	0	212
25	G12	0		60677	-1	-31	-4	-102	0	212
26	G13	0		60677	-18	-72	-60	-242	0	212
27	G14	0		60677	5	-43	17	-144	0	212
28	G15	0		60677	-5	-42	-16	-140	0	212
29	G16	0		60677	-9	-96	-30	-320	0	212

Figure 4-4 Orbit and Clock Day-Boundary Discontinuities in the Alignment Log File

For the dual-frequency code bias and wide-lane day-boundary discontinuities, the columns from left to right are: satellite PRN; frequency index (0 denotes a combination formed from the two reference frequencies); Modified Julian Date (MJD); geometry-free phase bias day-boundary discontinuity (ps); reference-channel differential code bias (DCB) day-boundary discontinuity (ps); code bias day-boundary alignment correction (expected to be 0); residual reference-channel DCB discontinuity after alignment (ps); wide-lane discontinuity (milli-cycles, mcy); wide-lane day-boundary alignment correction (mcy); and residual wide-lane discontinuity after alignment (mcy).

108 2. RESULTS OF DUAL-FREQUENCY GEOMETRY-FREE BIAS ALIGNMENT

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PRN	FRQ	A	MJDAY	PH_GFB (ps)	CD_GFB (ps)	CD_CORR (ps)	CD_RESI (ps)	WL_UPD (mcy)	WL_CORR (mcy)	WL_RESI (mcy)
G02	0	60677	-191	227	0	227	14	-22	-7	
G03	0	60677	-105	104	0	104	26	-22	4	
G04	0	60677	69	-41	0	-41	21	-22	0	
G05	0	60677	93	-131	0	-131	-974	978	4	
G06	0	60677	247	-235	0	-235	21	-22	0	
G07	0	60677	74	-70	0	-70	26	-22	4	
G08	0	60677	67	-18	0	-18	17	-22	-4	
G09	0	60677	-128	136	0	136	20	-22	-1	
G10	0	60677	9	15	0	15	24	-22	2	
G11	0	60677	182	-171	0	-171	22	-22	0	
G12	0	60677	115	-170	0	-170	-979	978	-1	
G13	0	60677	230	-245	0	-245	-975	978	3	
G14	0	60677	-98	73	0	73	18	-22	-3	
G15	0	60677	109	-154	0	-154	-980	978	-1	
G16	0	60677	-9	33	0	33	12	-22	-10	
G17	0	60677	-16	38	0	38	20	-22	-1	
G18	0	60677	14	-31	0	-31	27	-22	5	
G19	0	60677	157	-165	0	-165	22	-22	1	

Figure 4-5 Dual-Frequency Pseudorange-Bias and Wide-Lane Day-Boundary Discontinuities in the Alignment Log File

The narrow-lane phase bias day-boundary discontinuities are illustrated below. From left to right, the columns are: satellite PRN; frequency; MJD; narrow-lane phase bias day-boundary discontinuity; narrow-lane phase bias discontinuity after accounting for the orbit/clock alignment residuals; narrow-lane phase bias day-boundary alignment correction; and residual narrow-lane phase bias discontinuity after accounting for the orbit/clock alignment residuals. The final four quantities are each reported twice, once in picoseconds and once in milli-cycles.

203 3. RESULTS OF DUAL-FREQUENCY IRC ALIGNMENT

PRN	FRQ	A	MJDAY	NL_UPD (ps)	IRC (ps)	NL_CORR (ps)	IRC_RESI (ps)	NL_UPD (mcy)	IRC (mcy)	NL_CORR (mcy)	IRC_RESI (mcy)
208	G02	0	60677	-4	-89	-89	0	-11	-249	-249	0
209	G03	0	60677	-253	269	268	1	-708	755	751	4
210	G04	0	60677	33	-84	-89	5	93	-235	-249	14
211	G05	0	60677	-14	86	100	-14	-39	241	281	-40
212	G06	0	60677	65	-80	-89	8	183	-225	-249	24
213	G07	0	60677	93	-85	-89	4	261	-237	-249	12
214	G08	0	60677	-21	-95	-89	-7	-58	-267	-249	-19
215	G09	0	60677	69	-93	-89	-4	193	-261	-249	-12
216	G10	0	60677	47	-88	-89	1	133	-246	-249	2
217	G11	0	60677	67	-82	-89	7	188	-229	-249	20
218	G12	0	60677	6	107	100	7	17	301	281	20
219	G13	0	60677	296	-266	-257	-9	830	-745	-719	-26
220	G14	0	60677	131	-80	-89	9	366	-223	-249	25
221	G15	0	60677	-14	102	100	1	-38	285	281	4
222	G16	0	60677	14	-91	-89	-2	38	-254	-249	-5
223	G17	0	60677	41	-85	-89	4	116	-238	-249	11
224	G18	0	60677	139	-89	-89	-1	388	-250	-249	-2
225	G19	0	60677	-250	276	268	8	-701	775	751	24
226	G20	0	60677	76	-90	-89	-2	213	-253	-249	-5
227	G21	0	60677	-74	80	79	1	-208	225	222	3
228	G23	0	60677	-61	97	100	-3	-171	273	281	-8
229	G24	0	60677	161	-76	-89	13	450	-212	-249	36
230	G25	0	60677	-243	270	268	2	-680	757	751	6

Figure 4-6 Narrow-Lane Phase Bias Day-Boundary Discontinuities in the Alignment Log File

The code bias and phase bias alignment corrections are illustrated below. From left to right, the columns are: satellite PRN; frequency; MJD; code bias correction (ps); and phase bias correction (ps).

298 4. CORRECTIONS FOR CODE/PHASE BIASES

PRN	FRQ	A	MJDAY	OCB_CORR (ps)	OPB_CORR (ps)
303	G02	1	60677	0	-110
304	G03	1	60677	0	525
305	G04	1	60677	0	-110
306	G05	1	60677	0	-2014
307	G06	1	60677	0	-110
308	G07	1	60677	0	-110
309	G08	1	60677	0	-110
310	G09	1	60677	0	-110
311	G10	1	60677	0	-110
312	G11	1	60677	0	-110
313	G12	1	60677	0	-2014

Figure 4-7 Corrections Applied to Observable Biases in the Alignment Log File

Precise Orbit, Clock and Bias Product Files

The precise product files cover at least two consecutive days. The original input products remain unchanged. The target-day clock and bias products are aligned to the corresponding products of the preceding day. When orbit or clock extrapolation is performed, a corresponding “_predicted” SP3 or CLK file is generated with the extrapolated records appended after the last original epoch. When Write DOCB is set to YES, the precise bias files contain DOCB records, which quantify the residual discontinuities at day boundaries. The format is illustrated below.

72 +SOLUTION/DAY_BOUNDARY_DISCONTINUITY

*OBD	SVN	PRN	STATION	OBS1	OBS2	MIDNIGHT_AT	MIDNIGHT_AT	UNIT	ESTIMATED_VALUE	STD_DEV
74	DOCB	G061	G02	C1C		2025:001:86400	2025:002:00000	ns	0.04127	0.00000
75	DOCB	G061	G02	C1W		2025:001:86400	2025:002:00000	ns	0.03707	0.00000
76	DOCB	G061	G02	C2W		2025:001:86400	2025:002:00000	ns	0.00122	0.00000
77	DOCB	G061	G02	L1C		2025:001:86400	2025:002:00000	ns	-0.00009	0.00000
78	DOCB	G061	G02	L1W		2025:001:86400	2025:002:00000	ns	-0.00009	0.00000
79	DOCB	G061	G02	L2W		2025:001:86400	2025:002:00000	ns	-0.00009	0.00000
80	DOCB	G069	G03	C1C		2025:001:86400	2025:002:00000	ns	-0.00697	0.00000
81	DOCB	G069	G03	C1W		2025:001:86400	2025:002:00000	ns	-0.01487	0.00000
82	DOCB	G069	G03	C2X		2025:001:86400	2025:002:00000	ns	0.01083	0.00000
83	DOCB	G069	G03	C2S		2025:001:86400	2025:002:00000	ns	-0.02337	0.00000
84	DOCB	G069	G03	C2L		2025:001:86400	2025:002:00000	ns	-0.01467	0.00000
85	DOCB	G069	G03	C2W		2025:001:86400	2025:002:00000	ns	-0.01357	0.00000
86	DOCB	G069	G03	L1C		2025:001:86400	2025:002:00000	ns	-0.00148	0.00000
87	DOCB	G069	G03	L1W		2025:001:86400	2025:002:00000	ns	-0.00148	0.00000
88	DOCB	G069	G03	L2X		2025:001:86400	2025:002:00000	ns	-0.00148	0.00000
89	DOCB	G069	G03	L2S		2025:001:86400	2025:002:00000	ns	-0.00148	0.00000

Figure 4-8 DOCB Output Format

4.3.2 Usage of result data processing scripts/programs

Note

As shown in Table 4-2, there are some useful scripts under the “scripts/” directory that help users process. Note that if your Python 3 path is not “/usr/bin/python3”, you can use the **which** command to find the path of python3 in your computer.

```
which python3
/usr/local/bin/python3
```

Then, modify these python programs' header to the corresponding path.

```
#!/usr/bin/env python3
#!/usr/local/bin/python3
```

Copy the modified python scripts to the path where software environment variables are located (~/.PRIDE_DBAlign_BIN/). Then you can use these python scripts directly in any directory.

```
cp plot-*.py ~/.PRIDE_DBAlign_BIN/
```

Moreover, the Python plotting scripts require the Matplotlib packages.

Usage of **pdba_plot**

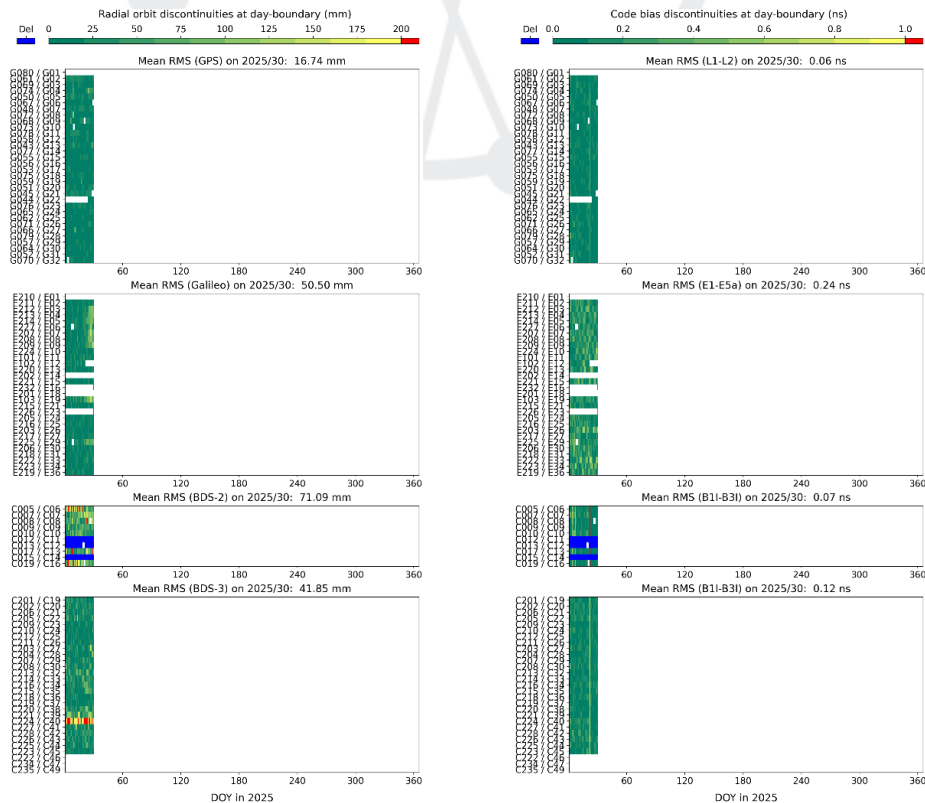
pdba_plot is used to generate heat maps of the residual day-boundary discontinuities after product alignment. Its usage is:

```
pdba_plot aligned_filefolder
```

where

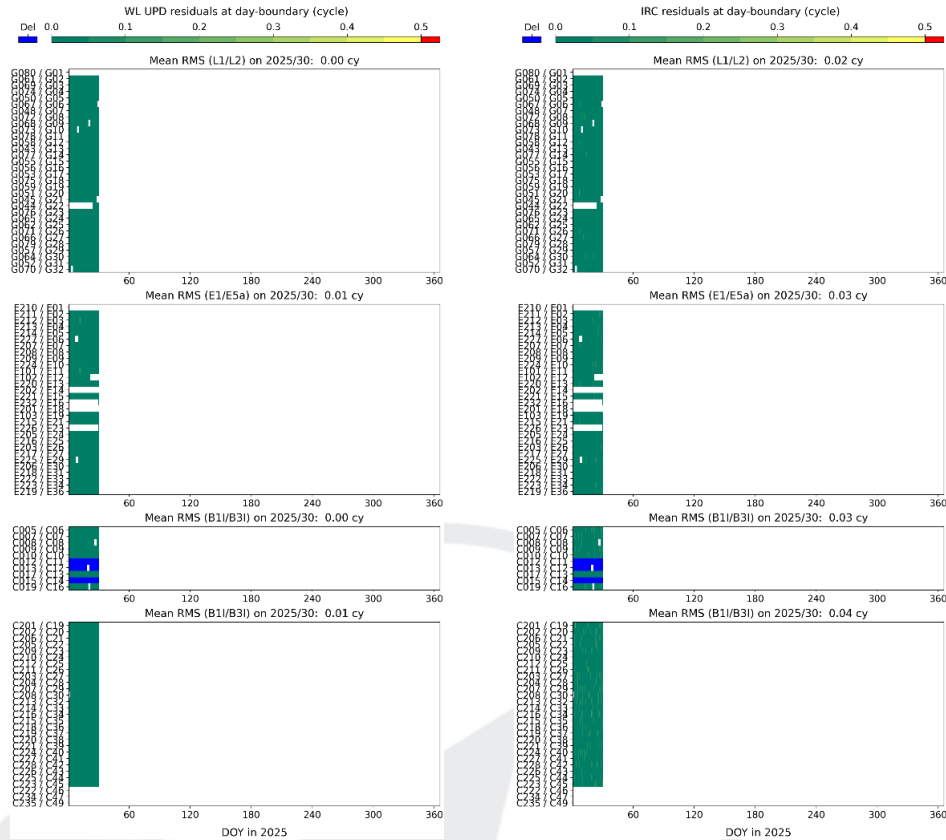
aligned_filefolder Directory containing the logs and products for the aligned day boundaries.

An example of the **pdba_plot** output is shown below, using the alignment logs for WUM Rapid products from day of year (DOY) 001 to 030 in 2025.



(a) Radial-orbit day-boundary discontinuities

(b) Code bias day-boundary discontinuities



(c) WL day-boundary alignment residuals (d) NL phase bias day-boundary alignment residuals

Figure 4-9 Example Output from `pdba_plot`

4.4 Quick Start and Program Execution

After the software installation is completed, the executable programs, processing scripts, configuration template, plotting programs, and satellite parameter file are installed in the directory `~/.PRIDE_DBAlign_BIN/`. The installer also adds this directory to the system PATH. On Linux, the PATH setting is written to `~/.bashrc`; on macOS, it is written to `~/.bash_profile`. Users should open a new terminal or reload the corresponding shell configuration file before running the software.

PRIDE DBAlign uses `algirc` as its core day-boundary alignment program. Although `algirc` is driven by a configuration file, the `pdba` command provides a simplified command-line interface. It automatically determines the processing date from the input filenames, locates the corresponding preceding day products, generates an ALGIRC configuration file, and invokes `algirc` to perform the alignment.

In this section, the software installation directory is denoted by `path-to-software`, the project working directory by `path-to-project`, the configuration file by `config_file`. All commands below are executed in `path-to-project`.

4.4.1 Usage of *pdba*

The usage of *pdba* is shown in Figure 4-10. For more detailed information, you can enter *pdba* in the terminal.

```

Usage: pdba -clk <file> -bia <file> -sp3 <file> [options]

Align the target-day clock and bias products to the previous day's original
products using the corresponding SP3 orbits. The target date is read from
the product filenames.

Start up:
-V, -v, --version          display version information
-H, -h, --help             display this help

Product options:
-clk <file>, --clock <file> target-day clock product (required)
-bia <file>, --bias <file>  target-day code/phase bias product (required)
-sp3 <file>, --sp3 <file>   target-day SP3 orbit product (required)

Processing options:
-wdocb <YES|NO>            write day-boundary discontinuity records
-cfg <file>, --config <file> base ALGIRC configuration file
-sys <char>, --system <char> GNSS systems selected from G, E and C
                           (default without -cfg: GEC)
-ubd23 <YES|NO>, --unify-bds23 <YES|NO> unify BDS-2/3 datum
                           (Default: YES for MJD >= 61152, otherwise NO)

Configuration precedence:
command-line option > configuration file > built-in default

```

Figure 4-10 pdba script partial help information

Except for “-clk”, “-sp3” and “-bia”, all other options are optional. The order of precedence is Command Line Parameters > Configuration File > Built in Default Value. The “-clk”, “-sp3” and “-bia” options must always be specified on the command line and are never read from the configuration file. Option names must be entered exactly as defined, while the values of “-wdocb”, “-sys” and “-ubd23” may be entered in either uppercase or lowercase. In the command syntax, < > indicates a required parameter and [] indicates an optional parameter. The options can be divided into three categories: (1) Mandatory product options; (2) Optional product and processing options; and (3) Start-up options. The meaning of each option is described below:

(1) Mandatory product options:

- **-clk <file>, --clock <file>**: Target day satellite clock product. This option is required.
- **-bia <file>, --bias <file>**: Target day code/phase bias product. This option is required.
- **-sp3 <file>, --sp3 <file>**: Target day satellite orbit product. This option is required.

(2) Optional product and processing options:

- **-cfg <file>, --config <file>**: Optional base ALGIRC configuration file. Either a relative or an absolute path may be used. A relative path is interpreted with respect to the current working directory. If omitted, the default configuration installed with *pdba* is used.
- **-sys <char>, --system <char>**: GNSS systems to be processed. One or more characters may be selected from G, E, and C, corresponding to GPS, Galileo, and BDS. For example, -sys GE processes GPS and Galileo. When neither -cfg nor -sys is specified, the default is GEC.
- **-ubd23 <YES|NO>, --unify-bds23 <YES|NO>**: Controls whether the BDS-2 and BDS-3 clock datums are unified. The value is case-insensitive. When neither -cfg nor -ubd23 is specified, the default is NO for MJD earlier than 61152 and YES otherwise. MJD 61152 (April 22, 2026) marks the completion of the BDS reconfiguration considered here.
- **-wdocb <YES|NO>**: Controls whether DOCB records are written to the BIA file. The value is case-insensitive. The default is YES.

(3) Start-up options:

- **-V, -v, --version**: Displays the version information of *pdba*.
- **-H, -h, --help**: Displays the command-line help information.

Note: *pdba* processes only one day boundary in each invocation and performs forward alignment only. For a target day D, the original clock, orbit, and bias products for both day D and day D-1

must be placed in the same directory. This requirement also applies when the two days cross a calendar-year boundary. *pdba* determines the processing date from the product filenames. For multi-day processing, run *pdba* separately for each target day.

4.4.2 Configuration File

The configuration template is located at `table/config_template` and uses the “Keyword = Value” format; text following “!” is treated as a comment, while lines beginning with # or * are treated as comment lines. The satellite list is defined as a separate block and does not use the Keyword = Value format.

In the current version of *pdba*, product files are specified primarily through command-line options rather than filename templates in the base configuration. The target day clock, orbit and bias products must be supplied using `-clk`, `-sp3` and `-bia`, respectively. *pdba* extracts the date from the target filenames, changes only the detected date field to construct the expected preceding day filenames, and searches for those files in the same product directory. It then generates a concrete configuration named `config_pdba_YYYYDDD` in that directory.

During this process, *pdba* overwrites Product directory, Satellite orbit, Satellite clock, Code/phase bias. The base configuration is mainly used to provide Write DOCB, Unify BDS-2/3, and the satellite list, subject to command-line options.

Product Paths and File Names

- Product directory = <product directory>

pdba derives the product directory from the target clock file. The target day clock, bias, and orbit products must be located in the same directory. The corresponding preceding day products must also be present in that directory. The generated configuration, ALGIRC log, and aligned products are written there.

- Satellite orbit = <preceding day SP3> <target day SP3>

The target day orbit file is supplied using `-sp3`. *pdba* constructs and locates the preceding day filename by replacing the detected date field in the target filename.

- Satellite clock = <preceding day CLK> <target day CLK>

The target day clock file is supplied using `-clk`. *pdba* constructs and locates the preceding day filename by replacing the detected date field in the target filename.

- Code/phase bias = <preceding day BIA> <target day BIA>

The target day OSB file is supplied using `-bia`. The corresponding preceding day file is located using the same date-field replacement rule as for the clock product.

- ERP file = <preceding day ERP>

The preceding day ERP file is used for coordinate transformations when the preceding day orbit product lacks the 24:00 epoch and orbit extrapolation is required. When `-cfg` is not specified, *pdba* automatically searches for this file in the preceding day SP3 directory. If no ERP file is available, the simplified coordinate transformation is used instead.

Alignment Options and Satellite List

- Write DOCB = YES

Write DOCB controls whether DOCB (Discontinuity of Orbits, Clocks, and Biases) records are written to the output BIA products. It should normally remain set to YES to support subsequent quality checks.

- **Unify BDS-2/3** = Default

Unify BDS-2/3 specifies whether BDS-2 and BDS-3 are treated as a unified system. The default is NO for MJD earlier than 61152 and YES otherwise.

- **+GNSS satellites / *PRN / -GNSS satellites**

This section defines the satellites included in the processing. Prefixing an individual satellite line with # disables that satellite.

4.4.3 General Operation Steps

After installation, consecutive daily products can be aligned as follows:

1. Confirm that install.sh has been executed successfully and that pdba can be invoked directly in a new terminal session.
2. Prepare uncompressed CLK, SP3, and BIA products for two consecutive days in the same product directory. Both the preceding day and target day products must be available.
3. Run pdba for the target day. For example:

```
pdba \
-cclk WUM0MGXRAP_20260020000_01D_30S_CLK.CLK \
-bia WUM0MGXRAP_20260020000_01D_01D_OSB.BIA \
-sp3 WUM0MGXRAP_20260020000_01D_05M_ORB.SP3 \
-ubd23 NO
```

For multi-day processing, run pdba separately for each target day.

4. Inspect the product directory. It contains the generated config_pdba_YYYYDDD configuration, the algirc-MJD.out log, and the aligned clock and bias products (and any “_predicted” SP3 or CLK files generated by extrapolation).
5. To examine the residual discontinuities at day boundaries, run:

```
pdba_plot path-to-products
```

pdba_plot requires Python and Matplotlib. It generates algirc-sum.out and the orbit, wide-lane, DCB, and IRC day-boundary residual plots in the specified directory.

4.4.4 Processing Example

The example/wum0rap directory provides examples based on WUM Rapid products. It contains SP3, CLK, and BIA products for DOY 001 and 002 of both 2025 and 2026. Because each alignment requires the preceding day products, DOY 002 is aligned using DOY 001 as the reference day.

Example: Day-Boundary Alignment of WUM Rapid Products

1. Change to the example directory:

```
cd path-to-software/example
```

2. Confirm that the required products are available in:

```
wum0rap/2025
```

```
wum0rap/2026
```

3. Run the example script:

```
./test.sh
```

The script performs the DOY 002 alignment for both 2025 and 2026 using the -clk, -bia, -sp3, and -ubd23 options.

4. Inspect the generated configurations, ALGIRC logs, and aligned products in the corresponding

year directories. The results can be plotted using:

```
pdba_plot wum0rap/2025
```

```
pdba_plot wum0rap/2026
```

To process products from another analysis center, provide the corresponding target day CLK, BIA, and SP3 filenames. The filenames must contain a supported date field, and the preceding day products must follow the same naming convention and be located in the same directory.



Appendix A. Instruction on Day-Boundary Aligned Products

This guide is intended for PPP users who want to apply aligned products (including orbits, clocks, and biases) to PPP software other than PRIDE PPP-AR. Users must be able to modify their software accordingly.

Following day-boundary alignment of the precise satellite products, clock estimates at 24:00:00 on the preceding day (the same boundary epoch as 00:00:00 on the target day) are provided for each GPS, Galileo, and BDS satellite in the CLK files. In addition, a numerical continuity indicator at this epoch, referred to as the DOCB (Discontinuity of Orbits, Clocks and Biases), is defined and provided in the Bias-SINEX files. To ensure optimal continuous data processing, users' PPP software must:

Read the estimates for the 24:00:00 epoch from the preceding day's product and the 00:00:00 epoch from the target day's product. Prevent orbit and clock interpolations across day boundaries (i.e., 24:00:00 or 00:00:00). This mechanism is illustrated in the following diagram. Taking orbit interpolation as an example, conventional methods (left) tend to place the interpolation point in the middle of the interpolation interval. However, in our proposed method (right), when the interpolation point falls on the preceding day, the epoch 24:00:00 of the preceding day's products should be used; when the interpolation point falls on the target day, the epoch 00:00:00 of the target day's products should be used instead. Because the interpolation methods for orbit and clock differ, the compatibility of their interpolation results will be compromised by the discontinuities in orbits and clocks. However, this problem can be avoided by interpolating them within a day.

